#1

Review.

events $\qquad$ RV. $\qquad$ Real numbers

$$\Omega \to \boxed{X(\cdot)} \to \mathbb{R}$$

$\{\omega_1, \omega_2 \cdots\}$ $\qquad$ $\{x_1, x_2 \cdots\}$

$\notin \mathbb{R}$

$E[X]$

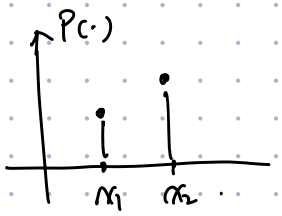
$Var[X]$

observations.

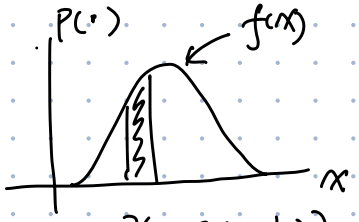
fixed values

realized values.

PMF



$P(X = x_1)$



$P(X \in (a,b))$

$= \int_a^b f(x)\, dx$

CDF: $P(X \le x) = F(x)$

$\sum_{x_i \le x} P(X = x_i)$ $\qquad$ $\int_{-\infty}^{x} f(t)\, dt$

$E[X] = \int x\, dF(x) = \mu$

$Var[X] = \int (x-\mu)^2\, dF(x) = E[X^2] - \{E[X]\}^2$

$= \int x^2\, dF(x) - \mu^2$

Riemman integral : $\int_R x f(x)\, dx$

Riemman – Stieltjes : $\int_R x\, dF(x)$ — $F(x) = P(X \le x)$

Lebesgue. : $\int_\Omega X(\omega)\, dP(\omega)$

Moment. $\quad E[X^k] = \int x^k\, dF(x)$

MGF: $\quad E\{e^{tX}\} = \int e^{tx}\, dF(x) = M_X(t)$

$$\frac{dM_X(t)}{dt} = \frac{d}{dt} \int e^{tx}\, dF(x) = \int \frac{d}{dt} e^{tx}\, dF(x)$$

$$= \int e^{tx} \cdot (x)\, dF(x) \overset{t=0}{\Rightarrow} \int x\, dF(x) = E[X]$$

RV: $X \sim f_X(x)$

Random sample: $X_1, X_2 \dots X_n \sim f_X(x)$

(IID): $X_i \perp X_j \quad \forall X_i \sim f_X(x)$

$f(x|\theta) = f(x;\theta)$

a function containing both Data (x) and parameter (θ)

PDF / PMF. $f(x|\theta)$ as a function of x

likelihood $f(x|\theta) = L(\theta|x)$ as a function of θ.

likelihood of random sample $L(\theta|x_1, x_2 \dots x_n) = f(x_1 \dots x_n|\theta) \overset{IID}{=} \prod_{i=1}^{n} f(x_i|\theta)$

Joint distribution

Exponential family.

$$f(x|\theta) = \underbrace{h(x)}_{\geq 0} \underbrace{c(\theta)}_{\geq 0} \exp\left\{ \sum_{i=1}^{k} \underbrace{w_i(\theta)}_{\text{function of parameter}} \underbrace{t_i(x)}_{\text{function containing only } x} \right\}$$

← pdf / pmf.

θ: vector. $\theta = (\theta_1 \dots \theta_d)$

usually $d = k$.

$$= h(x) \exp\left\{ \underbrace{\sum_{i=1}^{k} w_i(\theta) t_i(x)}_{W(\theta)^T T(x)} - A(\theta) \right\}$$

$$W(\theta)^T T(x) - A(\theta)$$

$$W(\theta)^T T(x) + q(\theta)$$

$$[w_1(\theta), w_2(\theta) \dots w_k(\theta)] \begin{bmatrix} t_1(\theta) \\ \vdots \\ t_k(\theta) \end{bmatrix}$$

log normalizer.

$$f(x|\theta) = \exp\left\{ \sum_{i=1}^{k} w_i(\theta)\, t_i(x) + d(\theta) + S(x) \right\}$$

natural parameter

Sufficient Statistics (Later)

$e^{S(x)} = h(x)$ base measure.

$\theta = g(\mu)$ g: link function.

canonical link.

Given ϕ is known.

outcome variable.

GLM $$f(y|\theta) = \exp\left\{ \frac{y\theta - b(\theta)}{a(\phi)} - c(y,\phi) \right\}$$

dispersion parameter

$$= \exp\left\{ y \cdot \frac{\theta}{a(\phi)} - \frac{b(\theta)}{a(\phi)} - c(y,\phi) \right\}$$

$$= \exp\left\{ y \cdot c(\tilde{\theta}) + d(\tilde{\theta}) + S(y) \right\}$$

$$= \exp\left\{ T(y)\, c(\tilde{\theta}) + d(\tilde{\theta}) + S(y) \right\}$$

$w_i(\theta) = g(\theta)$

link function.

LR: $y \sim N(\mu, \sigma^2)$

$g(\mu) = \mu = \alpha + \beta X$.

Logistic regression $y \sim$ Bernoulli

$g(p) = \log\left(\frac{p}{1-p}\right) = \alpha + \beta X$ $p = P(Y=1)$

Poisson regression

$g(\lambda) = \log(\lambda) = \alpha + \beta X$

(3.4.7). $f(x|\eta) = h(x)\, \underline{\underline{c^*(\eta)}} \exp\left\{ \sum_{i=1}^{k} \underline{\eta_i}\, t_i(x) \right\}$

pdf / pmf

$w_i(\theta)$

$w_i(\theta_1, \theta_2, \dots \theta_d)$

$w_1(\theta_1 \cdots \theta_d) \to \eta_1$

$\vdots \qquad \vdots$

$w_k(\theta_1 \cdots \theta_d) \to \eta_k$

$d<k \Rightarrow$ curved exponential family.

$d=k \Rightarrow$ full exponential family.

natural parameter space.

Exist.

$$\mathcal{H} = \left\{ \eta = (\eta_1 \cdots \eta_k) : \int_{-\infty}^{\infty} h(x) \exp\left(\sum \underline{\underline{\eta_i}}\, t_i(x) \right) dx < \infty \right\}$$

$$\sum h_i(x) \exp\{ \sum __) < \infty$$

$$c^*(\eta) = \left\{ \int_{-\infty}^{\infty} h(x) \exp\left\{ \sum_{i=1}^{k} \eta_i\, t_i(x) \right\} dx \right\}^{-1}$$

sufficient statistics

Reexpress of 3.4.7. $f(x|\eta) = \dfrac{1}{\{\Delta\}} h(x) \exp\{ \eta^T T(x) \}$ $\qquad (\eta_1, \eta_2 \cdots \eta_k) \begin{pmatrix} t_1(x) \\ \vdots \\ t_n(x) \end{pmatrix} = \eta^T T(x)$

natural parameter.

containing link function

$\downarrow c^*$

cont: $\int h(x) \exp\{ \quad \} dx$

disc. $\sum h(x) \exp\{ \quad \} dx$

Bernoulli Distribution.

$$f(x) = \begin{cases} p & \text{if } x=1 \\ 1-p & \text{if } x=0 \end{cases} \iff f(x|p) = p^x (1-p)^{1-x}$$

$$= (1-p)\cdot\left(\frac{p}{1-p}\right)^x$$

link function $\underline{\underline{\log\frac{x}{1-x}}}$.

$$= \underbrace{(1-p)}_{c^*(p)} \cdot \underbrace{1}_{h(x)} \cdot \exp\Big\{ \underbrace{\left(\log\frac{p}{1-p}\right)}_{\eta} \cdot \underbrace{x}_{T(x)} \Big\}$$

$$\Downarrow$$

$$c^*(p) = \frac{1}{\sum_{x=0,1} \exp\{x \cdot \log\frac{p}{1-p}\}}$$

$$= \frac{1}{\left(\frac{p}{1-p}\right)^0 + \left(\frac{p}{1-p}\right)^1} = \frac{1}{1+\frac{p}{1-p}} = \frac{1-p}{1-p+p} = \underline{\underline{1-p}}.$$

Binomial Dist. (3.4.1)

$$f(x|p) = \binom{n}{x}(1-p)^n \exp\Big\{\log\frac{p}{1-p}\cdot x\Big\}$$

$$= \binom{n}{x} \exp\Big\{ \underbrace{\log\frac{p}{1-p}}_{\eta} \cdot \underbrace{x}_{T(x)} + \underbrace{n\log(1-p)}_{A(\eta)} \Big\}.$$

$\downarrow$ link function : $\log\frac{p}{1-p}$